

Ayush Sunil Munot

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EDUCATION

Indian Institute of Technology Kharagpur, India

2021 – 2026

B.Tech. in Mechanical. Engg., M.Tech. in Financial Engg. (5Y); Minor: CSE; Micro Spl.: AI

CGPA: 8.95 / 10

PUBLICATIONS

MAEB: Massive Audio Embedding Benchmark | Under Review, *NeurIPS 2026* | *arXiv*

DiffusedDINO: Fine-Grained Visual Perception for Puzzle Solving via DINO | Under Review, *NeurIPS 2026*

EXPERIENCE

AI Intern | *Fidelity Investments* | Bangalore, India

May 2025 – July 2025

- Built a retrieval-based agent assist system for email/chat/call workflows, reducing query **resolution time** by **30%**
- Indexed **2300+** documents using **FAISS**, built **SQLite** feedback loops, and deployed via Streamlit + **FastAPI**
- Achieved **90%+** top-K retrieval accuracy using **HyQE**, outperforming RAG (**60%**) on a 10k-query test set

ML Engineering Intern | *Sourcegraph* | Remote

May 2024 – Jan 2025

- Improved multi-language unit test generation using **Dspy**, achieving **~10%** gains on **HumanEvalPack** dataset
- Built a **multi-agent** system to process **100k+** Cody logs, enabling dataset creation, fine-tuning, & evaluation
- Synthesized retrieval datasets, integrated **10+** code-retrieval benchmarks, evaluated Cody against **5+** competitors

Research Assistant | *AFAR Lab, University of Cambridge* | Remote

July 2023 – Jan 2024

- Constructed temporal dataset from **OpenFace** features, mapping to frequency-aligned **spectral** representations
- Transformed temporal data into spectral graphs for **GNN**-based multi-class classification with **early & late** fusion
- Achieved **85%** accuracy (**GAT**) & **55%** (**GCN**), surpassing **45%** ML baseline using **Optuna** for parameter tuning

Computer Vision Intern | *Spectrum Lab, IISc Bangalore* | Remote

May 2023 – Aug 2023

- Developed **STAPLECAM**, an **EM**-based endoscopic segmentation method integrating multiple CAM techniques
- Finetuned **ResNet-152** on endoscopic data with cross-entropy loss & **Otsu**-based binarization using 20% test split
- Achieved Dice/IoU scores of **0.86/0.92** on the KVASIR dataset & **0.74/0.85** on the BKAI-IGH-NeoPolyp dataset

PROJECTS

Multimodal AI Notes Platform (PWA) | *self-project* | *GitHub*

Feb 2026 – Mar 2026

- Built a multimodal **PWA** for note-taking using **FastAPI** & **Supabase**, supporting text, audio, & image ingestion
- Designed a pipeline with **Gemini**-based parsing, **pgvector** semantic search, & secure storage via signed URLs
- Shipped **Gemini 2.5 TTS** summaries, deep-link sharing, & **MCP** integrations for AI-assisted knowledge access

Multi-Agent AI for Interactive Physics Simulations | *self-project* | *GitHub*

July 2025 – Aug 2025

- Built a self-correcting **multi-agent** system for physics simulation generation, reducing prototyping time by **80%**
- Integrated code **sandbox** with automated dependency management, achieving **90%** autonomous execution rate
- Expanded into a **multimodal** learning platform with **context-aware** chat, voice/image inputs for faster iteration

Adobe Behaviour Simulation Challenge - Gold | *Inter IIT TECH Meet - 12.0*

Nov 2023 – Dec 2023

- Designed **MLP+Transformer** head on **BERT** with **SAM** losses & **PCA**-based inference for tweet-like prediction
- Built supervised routing to optimally select among **3** fine-tuned LLMs; used **CLIP** for enhanced tweet generation
- Achieved **2262.33 RMSE** predicting likes; surpassed **GPT-4** in content prediction (**0.24 BLEU**, **0.25 ROUGE**)

TECHNICAL SKILLS

Languages: C, C++, Python, SQL

ML & DL Frameworks: NumPy, Pandas, SciKit-learn, TensorFlow, Keras, PyTorch, OpenCV, Matplotlib, Optuna

Data & MLOps: FAISS, PostgreSQL, SQLite, Redis, Apache Spark, AWS, GCP, Docker, CI/CD, MLflow

Developer Tools: Git, GitHub, Bash, Linux, Streamlit, FastAPI, Pydantic, Gradio, VS Code

NLP, DL & GenAI: Deep Learning, Graph Neural Networks, Hugging Face, Transformers, Embeddings, Vector

Databases, LLM Fine-tuning & Inference, RAG, Agentic Systems, LangChain, LangGraph, MCP, Prompt Engineering

AWARDS & ACHIEVEMENTS

- Core Contributor** in **MTEB** (Massive Text Embedding Benchmark), a leading evaluation repository
- Served as a **Reviewer** for the **Industry Track of ACL 2026**, evaluating applied NLP research submissions
- Selected for **Amazon ML Summer School 2025**, a prestigious program on advances in machine learning
- Achieved top **0.4%** in JEE-Mains 2021 and top **4%** in JEE-Advanced 2021 among 2.5 Lakh candidates
- Guardian** on LeetCode (top **1.13%**), Max Rating **2211** (Ayush-1234); Global Rank **102** in Weekly Contest 432

COURSEWORK

Maths & CS: Linear Algebra, Probability & Statistics, Time Series, Data Structures & Algorithms, Complex Networks

Machine Learning: ML Foundations, NLP, Computer Vision; **MOOCs:** CS229, CS230, CS231n, CS224n, CS336